

genQE — Two-Nucleon GCF Event Generator

Physics & Pipeline Documentation

GCF with 3N Project

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Contents

1 Overview

genQE is a Monte Carlo event generator for quasi-elastic (QE) electron scattering off nuclei within the **Generalized Contact Formalism (GCF)**. It generates events of the type

$$e + A \longrightarrow e' + N_{\text{lead}} + N_{\text{rec}} + (A-2)^*, \quad (1)$$

where the electron scatters off one nucleon of a short-range correlated (SRC) pair. The struck nucleon (N_{lead}) carries the virtual-photon momentum; the partner (N_{rec}) recoils, and the residual ($A-2$) system is left with excitation energy E^* .

The generator factorizes the problem into two independent parts:

1. **Nuclear structure:** sampling the SRC pair's internal momenta via the GCF (contact coefficients $\times$ universal functions).
2. **Scattering dynamics:** computing the electron–nucleon cross section at the sampled kinematics.

2 Generalized Contact Formalism

2.1 Contact factorization

The GCF expresses the two-body momentum distribution of a nucleus A as

$$n_A^{(ij)}(\mathbf{k}_{\text{rel}}) = C_{ij}^A |\varphi_{ij}(k_{\text{rel}})|^2, \quad (2)$$

where i, j label isospin–spin channels, C_{ij}^A are the *nuclear contact coefficients* (nucleus-dependent, extracted from *ab initio* calculations or data), and $\varphi_{ij}(k)$ are *universal* (nucleus-independent) functions determined by the short-range behavior of the chosen NN potential.

Four channels are implemented:

Channel	Pair	Quantum numbers
pp0	proton–proton	1S_0
nn0	neutron–neutron	1S_0
pn0	proton–neutron	1S_0
pn1	proton–neutron	3S_1 (deuteron-like, dominant)

For identical-nucleon pairs (pp or nn) the spectral function used is

$$S(k) = 2 C \varphi^2(k), \quad (3)$$

while for pn pairs it is

$$S(k) = C_{pn0} \varphi_{pn0}^2(k) + C_{pn1} \varphi_{pn1}^2(k). \quad (4)$$

The universal functions $\varphi^2(k)$ are tabulated in 100 bins covering $k \in [0, 10] \text{ fm}^{-1}$ with $\Delta k = 0.1 \text{ fm}^{-1}$ and linearly interpolated at runtime.

2.2 Available NN interactions

Several sets of $\varphi^2(k)$ and contact values are included:

- **AV18** — Argonne v_{18} (default)
- **N2LO_10** — Chiral N^2LO , $R = 1.0 \text{ fm}$
- **N2LO_12** — Chiral N^2LO , $R = 1.2 \text{ fm}$
- **N3LO_600** — Chiral N^3LO , $\Lambda = 600 \text{ MeV}$
- **AV4Pc** — Argonne $\text{AV4}'$
- **NV2_1a** — Norfolk $\text{NV2}+\text{Ia}$
- **AV18_deut** — AV18 with deuteron-specific wave function (no contact dependence)

2.3 Contact coefficients

Contact values C_{ij}^A are stored per nucleus and NN interaction. They are loaded by `gcfSRC` on initialization. Representative values (AV18, ^{12}C):

$$C_{pp0} \approx C_{nn0} \approx 5.4, \quad C_{pn0} \approx 5.4, \quad C_{pn1} \approx 110.7. \quad (5)$$

The $pn1$ (deuteron-like) channel dominates by roughly an order of magnitude.

3 Nuclear Parameters

Each target nucleus is described by a `gcfNucleus` object that stores:

- A , Z , nuclear mass m_A
- σ_{CM} : width (GeV/ c) of the Gaussian center-of-mass momentum distribution of the pair
- E_{max}^* : maximum excitation energy of the $(A-2)$ residual
- $(A-2)$ masses: m_{A-2}^{pp} , m_{A-2}^{pn} , m_{A-2}^{nn} (depend on which pair is removed)

Nucleus	A	σ_{CM} (GeV/ c)	E_{max}^* (GeV)	Note
^2H	2	0	0	No CM motion, no residual
^3He	3	0.10	0	$(A-2)$ = single nucleon
^4He	4	0.10	0.010	$(A-2)$ = deuteron (pn) or $2N$ (pp/nm)
^{12}C	12	0.15	0.030	
^{56}Fe	56	0.15	0.050	
^{208}Pb	208	0.15	0.100	

The excitation energy can be sampled from a Gaussian $E^* \sim \mathcal{N}(E_{\text{mean}}^*, \sigma_E)$ or held fixed.

4 Event Generation Pipeline

4.1 Step 1: Pair type selection

A pair type $(ij) \in \{pp, nn, pn\}$ is chosen uniformly. The event weight carries an overall factor of 4 (2 nucleons $\times$ 2 possible isospin assignments).

4.2 Step 2: Decay function — sampling the SRC pair

The SRC pair's kinematics are generated in `gcfGenerator::decay_function()`.

Center-of-mass momentum. Each Cartesian component of $\mathbf{P}_{\text{CM}}$ is sampled independently from a Gaussian:

$$P_{\text{CM},\alpha} \sim \mathcal{N}(0, \sigma_{\text{CM}}). \quad (6)$$

Relative momentum. The relative momentum $\mathbf{k}_{\text{rel}}$ is sampled uniformly:

$$k \in [k_{\text{min}}, k_{\text{max}}], \quad \cos \theta_{\text{rel}} \in [-1, 1], \quad \phi_{\text{rel}} \in [0, 2\pi]. \quad (7)$$

Default range: $k_{\text{min}} = 0.2 \text{ GeV}/c$, $k_{\text{max}} = 1.05 \text{ GeV}/c$.

Weight from spectral function. The phase-space weight is

$$w_{\text{decay}} = \Delta k \Delta(\cos \theta) \Delta \phi \times \frac{k^2 S(k, i, j)}{(2\pi)^3}, \quad (8)$$

where $S(k)$ is the GCF spectral function from Eq. (??).

Pauli blocking cut. If $k < k_{\text{cut}}$ (default 0.25 GeV/c; 0 for deuterium), the event is rejected ($w = 0$).

Individual momenta.

$$\mathbf{p}_{\text{lead}} = \frac{1}{2}\mathbf{P}_{\text{CM}} + \mathbf{k}_{\text{rel}}, \quad \mathbf{p}_{\text{rec}} = \frac{1}{2}\mathbf{P}_{\text{CM}} - \mathbf{k}_{\text{rel}}. \quad (9)$$

Lead nucleon energy. From energy conservation:

$$E_{\text{lead}} = m_A - E_{A-2} - E_{\text{rec}}, \quad E_{\text{rec}} = \sqrt{|\mathbf{p}_{\text{rec}}|^2 + m_N^2}, \quad (10)$$

where $E_{A-2} = \sqrt{|\mathbf{p}_{A-2}|^2 + m_{A-2}^2}$ and $\mathbf{p}_{A-2} = -\mathbf{P}_{\text{CM}}$.

4.3 Step 3: Light-cone variant

An alternative light-cone kinematic mode (`decay_function_lc`) is available. It uses light-cone momentum fractions

$$\alpha = \frac{p^+}{\bar{m}}, \quad \bar{m} = m_A/A, \quad (11)$$

with transverse momentum $\mathbf{p}_\perp$. The relative light-cone momentum fraction α_{rel} is sampled uniformly, and the constraint

$$k_{\text{min}} = \frac{|1 - \alpha_{\text{rel}}|}{\sqrt{\alpha_{\text{rel}}(2 - \alpha_{\text{rel}})}} m_N \quad (12)$$

is enforced to keep kinematics physical. The weight is

$$w_{\text{lc}} = \Delta\alpha \Delta k \Delta\phi \times \frac{k \sqrt{k^2 + m_N^2} S(k)}{(2\pi)^3}. \quad (13)$$

4.4 Step 4: Electron scattering vertex

Given the lead nucleon's 4-momentum from Step 2, the scattered electron kinematics are determined.

Sampling. Two variables are drawn uniformly:

$$Q^2 \in [Q_{\text{min}}^2, Q_{\text{max}}^2], \quad \phi_k \in [0, 2\pi]. \quad (14)$$

Kinematic constraint. The electron's transverse momentum $k_\perp$ is determined by solving a quadratic arising from energy-momentum conservation of the full reaction. Let $p_1^- = E_1 - p_{1z}$ denote the nucleon's light-cone minus component and $k^- = Q^2/(2E_{\text{beam}})$. Then

$$A_e = \frac{k^-}{p_1^-}, \quad c_e = A_e(p_1^+ k^- - 2E_{\text{beam}} p_{\text{lead}}^- - \text{virt}), \quad b_e = -2A_e y, \quad (15)$$

where $y = p_{1\perp} \cos(\phi_k - \phi_1)$ and “virt” encodes off-shell effects. The solution is

$$k_\perp = \frac{-b_e \pm \sqrt{b_e^2 - 4c_e}}{2}. \quad (16)$$

Events with $D = b_e^2 - 4c_e < 0$ are rejected. When two physical solutions exist, both are equally probable and the weight is multiplied by 2.

Jacobian. The δ -function constraint contributes a Jacobian factor

$$J = 2 E_{\text{beam}} E_k \left| 1 - \frac{p_{1z} + y \tan(\theta_k/2) + E_{\text{beam}} - E_k}{E_{\text{lead}}} \right|, \quad w = 1/J. \quad (17)$$

4.5 Step 5: Cross section

The electron–nucleon cross section σ_{eN} is evaluated at the sampled kinematics. Three prescriptions are available:

On-shell. Standard Rosenbluth formula assuming the nucleon is on-shell:

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_{\text{beam}}^2 \sin^4(\theta/2)} \frac{E_k}{E_{\text{beam}}} \left[\frac{G_E^2 + \tau G_M^2}{1 + \tau} \cos^2 \frac{\theta}{2} + 2\tau G_M^2 \sin^2 \frac{\theta}{2} \right], \quad (18)$$

with $\tau = Q^2/(4m_N^2)$.

CC1 (de Forest, default). Off-shell generalization using current-conservation prescription 1. Constructs structure functions w_C, w_T, w_S, w_I from form factors G_E, G_M , and evaluates

$$\sigma_{\text{CC1}} = \sigma_{\text{Mott}} [v_L w_C + v_T w_T + v_S w_S + v_I w_I], \quad (19)$$

where $v_{L,T,S,I}$ are kinematic response functions.

CC2. Alternative current-conservation prescription with modified structure function weighting (different treatment of $\bar{p}$ components).

4.6 Step 6: Form factors

Three electromagnetic form-factor models:

Dipole. $G_E^p = G_D = (1 + Q^2/0.71)^{-2}$, $G_M^p = \mu_p G_D$.

Kelly (default). Rational parameterization:

$$G(Q^2) = \frac{1 + a_1 \tau}{1 + b_1 \tau + b_2 \tau^2 + b_3 \tau^3}. \quad (20)$$

Fit parameters given separately for $G_E^p, G_M^p/\mu_p, G_E^n, G_M^n/\mu_n$.

Ye. 13-parameter z -expansion fit.

4.7 Step 7: Final weight

The total event weight is

$$w = 4 \times w_{\text{decay}} \times (\Delta Q^2 \cdot 2\pi) \times \frac{1}{J} \times \sigma_{eN}. \quad (21)$$

4.8 Step 8: Optional corrections

Coulomb correction. The electron energy is shifted before and after scattering by the nuclear Coulomb potential:

$$\Delta E_{\text{Coulomb}} = \frac{3}{2} \frac{Z\alpha_{\text{em}}}{R_C}, \quad R_C = 1.1 A^{1/3} \text{ fm}. \quad (22)$$

Bremsstrahlung. Radiative corrections via the peaking approximation. A Poisson-distributed number of photons is emitted; each photon’s energy fraction sampled from

$$P(\Delta E/E) \propto \left(\frac{\Delta E}{E}\right)^{\lambda-1}, \quad \lambda = \frac{\alpha}{\pi} \left(\ln \frac{4E^2}{m_e^2} - 1\right). \quad (23)$$

5 Output

Events are stored in a ROOT `TTree` named `genT` with branches:

Branch	Type	Description
<code>lead_type</code>	int	PDG code (2212 = p , 2112 = n)
<code>rec_type</code>	int	PDG code of recoil nucleon
<code>pe[3]</code>	double	Scattered electron $\mathbf{p}_{e'}$
<code>pLead[3]</code>	double	Lead nucleon $\mathbf{p}_{\text{lead}}$
<code>pRec[3]</code>	double	Recoil nucleon $\mathbf{p}_{\text{rec}}$
<code>pAm2[3]</code>	double	$(A-2)$ momentum
<code>pRel[3]</code>	double	Relative momentum $\mathbf{k}_{\text{rel}}$
<code>q[3]</code>	double	Momentum transfer $\mathbf{q}$
<code>weight</code>	double	Event weight

6 Summary of Algorithm

For each Monte Carlo trial:

1. Select pair type (pp , nn , or pn) uniformly.
2. Sample $\mathbf{P}_{\text{CM}}$ (Gaussian) and $\mathbf{k}_{\text{rel}}$ (uniform in k , $\cos\theta$, ϕ).
3. Evaluate GCF spectral function $S(k)$; apply Pauli cut.
4. Derive individual nucleon momenta and $(A-2)$ recoil.
5. (Optional) Apply Coulomb shift and bremsstrahlung to beam.
6. Sample Q^2 and ϕ_k ; solve quadratic for $k_{\perp}$.
7. Compute Jacobian J from energy conservation constraint.
8. Evaluate σ_{eN} (CC1/CC2/on-shell with Kelly/dipole/Ye).
9. Multiply all weight factors; store event if $w > 0$.